

DEREK WANG

Canadian Citizen | (416) 898-6127 | derek.wang1@uwaterloo.ca | [in/4derekwang/](https://github.com/4derekwang/) | github.com/4ppleSA0CE | 4pplesa0ce.github.io

Education

University of Waterloo

Bachelor of Applied Science in Computer Engineering

Expected Apr 2030

GPA: 3.7

Experience

Autonomous Self Driving Software Engineer

WATonomous

Sept 2025 – Present

Waterloo, ON, Canada

- Coordinated collection and annotation of a comprehensive dataset of 3,448 images (3,087 train, 240 valid, 121 test) for object detection, implementing strategic data augmentation to ensure high-quality training data
- Trained YOLO11s object detection model using Ultralytics framework, achieving a 167% performance boost from 30% to 80% confidence rate through optimized hyperparameters and data augmentation strategies
- Exported trained model to ONNX format with onnxslim optimization for embedded deployment, creating a 36MB artifact ready for real-time inference on rover hardware

Autonomy Developer

Waterloo Aerial Robotics Group

Aug 2025 – Present

Waterloo, ON, Canada

- Developed Hardware-in-the-Loop (HITL) simulation systems in Python to address autonomous flight testing requirements for the 2026 National Annual UAS Student Competition
- Redesigned GPS movement architecture using periodic functions to eliminate unrealistic teleportation behavior, creating smooth trajectory interpolation systems that enhanced simulation realism and accuracy by 75%

Robotics Software Programmer – VEX Robotics Competition

Checkmate Robotics Club

Sept 2024 – June 2025

Markham, Ontario

- Programmed competition software for Team 16868C Rushdown in the VEX Robotics Competition, securing invitation to the 2025 VRC World Championships among top 10% of global teams
- Deployed advanced robotics systems using Okapi PROS Library, Odometry positioning, and PID Controllers, boosting robot performance by 25% through precise motion control and comprehensive error handling
- Designed and refactored autonomous navigation algorithms, creating 15-second match routes and 60-second programming skills routines that achieved 5th place out of 80 teams at Provincial Championships

Projects

Coffee Chat Scheduler | React | Python | FastAPI | PostgreSQL | Google Calendar API | Git | [Source Code](#) | [Live Demo](#)

- Built a full-stack web application using React and FastAPI, integrating Google Calendar API via OAuth 2.0 authentication with RESTful API endpoints, error handling, and CORS configuration
- Implemented real-time availability checking and automated calendar event creation, storing booking data in a PostgreSQL database with SQLAlchemy Object Relational Mapping

Autonomous Vehicle Control System | C++ | ROS2 | Docker | CMake | Foxglove | [Source Code](#) | [Demo Video](#)

- Engineered LiDAR-based perception pipeline processing sensor data to generate real-time costmaps with 0.1m resolution, integrating persistent memory mapping system for 95% accurate dynamic environment representation
- Architected a modular ROS2 navigation stack using A* pathfinding and Pure Pursuit control, achieving sub-0.5 m waypoint accuracy and 100% obstacle avoidance across 50+ scenarios.

Fruit Ninja AI | Python | YOLO11 | [Source Code](#)

- Architected and implemented an automated computer vision pipeline using YOLOv11 object detection, achieving 90% accuracy in fruit classification
- Built comprehensive dataset of 816 images (514 training, 302 testing), optimizing model performance through strategic data augmentation and validation techniques

Skills

Languages: C++, C, Java, Python, HTML5, CSS, JavaScript, C#, TypeScript, SQL

Engineering Tools: Git, Fusion 360, Docker, Foxglove, Unity, PostgreSQL, MongoDB, Supabase

Technologies: React, FastAPI, Node.js, Express, ROS2, PyTorch, OpenCV, Ultralytics, Tailwind CSS

Certifications: Docker Foundations Professional